Smart Energy System for Glassboro

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Venue: Room 225, Business Hall, Rowan University

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Introduction

1. Smart energy distribution is key to modern urban planning, aiming to optimize energy use and integrate renewable sources.
2. It enhances the **efficiency and reliability** of power grids through intelligent energy management systems.
3. The approach leverages **IoT and AWS technologies** to build a sustainable and effective energy framework for cities.

Reference: (https://www.energy.gov/oe/grid-modernization-and-smart-grid)

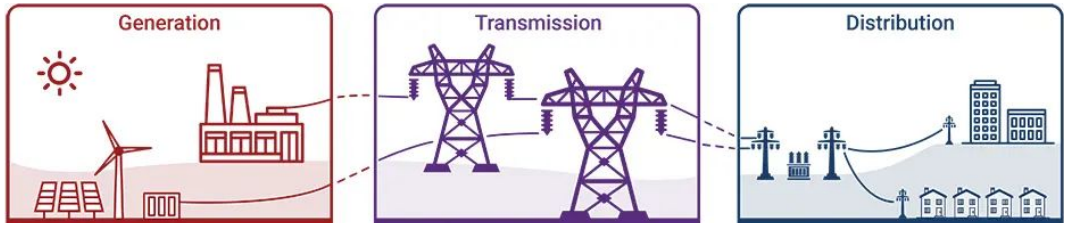
BACKGROUND

Aims and Objectives

Aim: Develop a smart microgrid using IoT monitoring system for the city of Glassboro & encourage energy efficiency and sustainability.

Objectives:

- Study the power grid existing in Glassboro.
- Identify the energy hot spots in Glassboro.
- Design a system of internet-connected sensors using AWS.
- Analyze performance & report on benefits of such connectivity.



(Reference: https://clearpath.org/tech-101/intro-to-energy-storage/)

Problem Statement

Glassboro is a modern city, however challenges with power distribution and non-efficient usage is no different from other modern cities.

These includes:

- **Wasteful Energy** usage due to non-usage of smart energy systems, tools and devices.
- As energy demand continues to grow, newer alternatives need to be introduced, like **integration of renewables and storage solutions**.
- Use of **non-efficient household and commercial devices and applications which needs control**.

Sectors of Glassboro, NJ

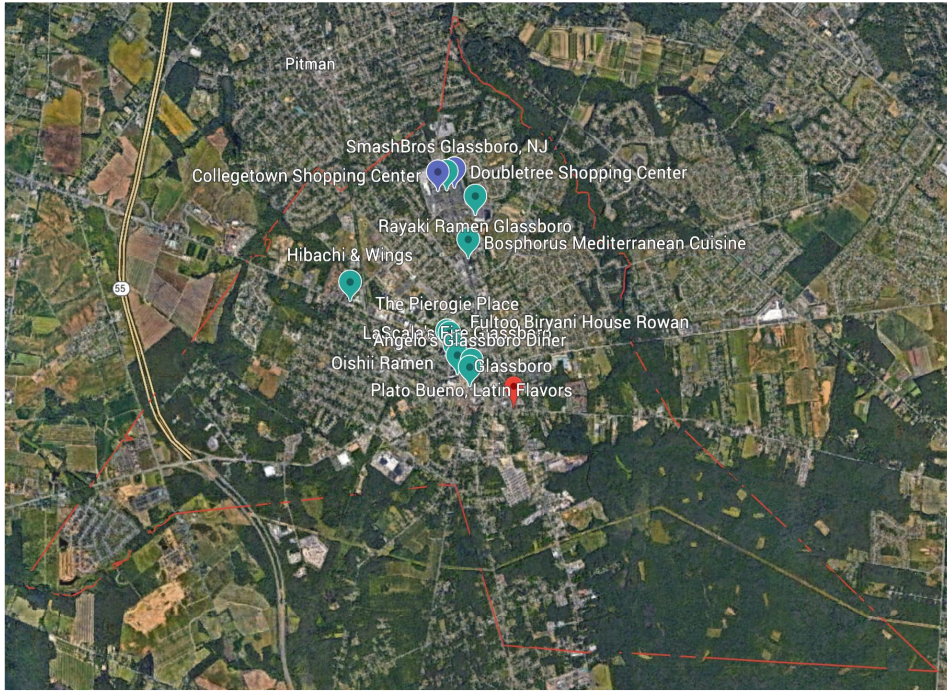


Figure 1: Commercial Sector of Glassboro

Figure 2: Residential Sector of Glassboro

https://earth.google.com/earth/d/1ODd81ypt00TR-odoUXNj8-uJtJ6g2668?usp=sharing

Sectors of Glassboro, NJ

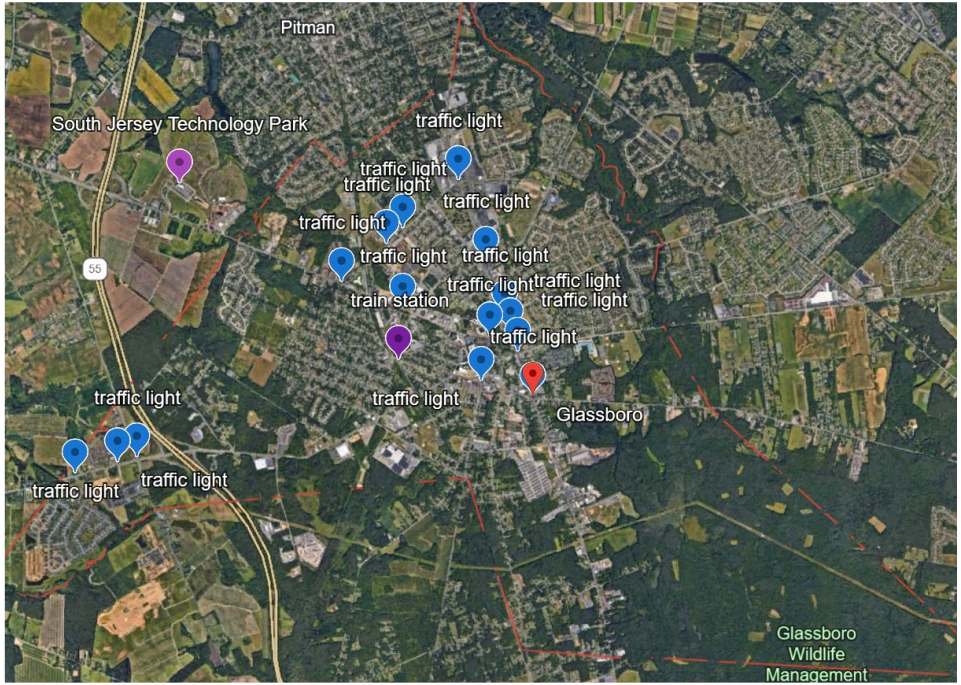


Figure 3: Transportation and Industrial Sectors of Glassboro

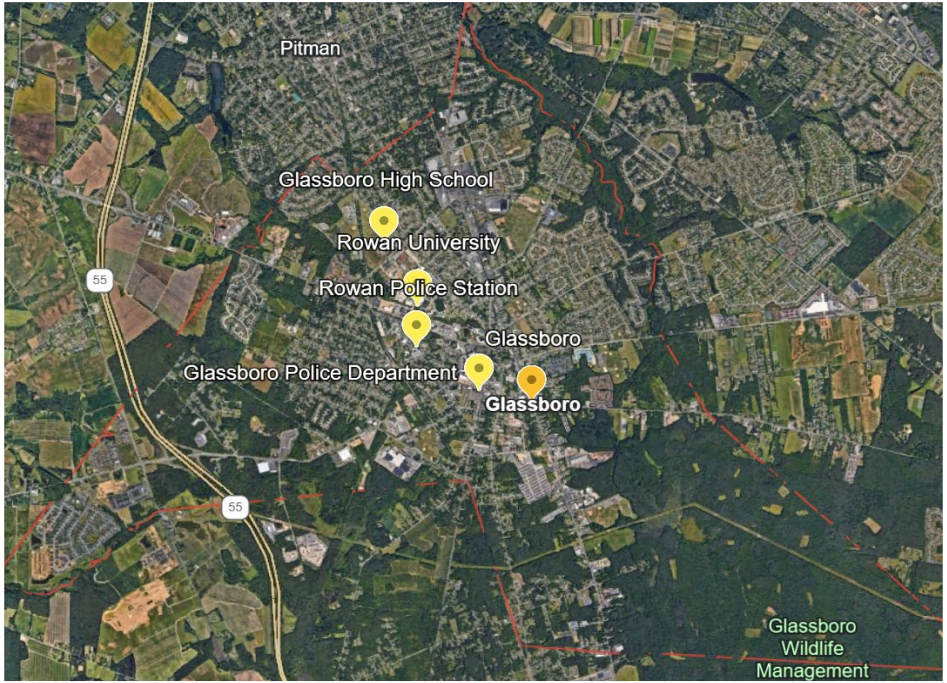


Figure 4: Public and Infrastructure Sector of Glassboro

https://earth.google.com/earth/d/1ODd81ypt00TR-odoUXNj8-uJtJ6g2668?usp=sharing

Average Energy Consumption in Glassboro:

Sector	# of Installations (Hypothetical)	Hourly Rate (kWh/hour)	Daily Rate (kWh/day)	Monthly Rate (kWh/month)	Yearly Rate (kWh/year)
Commercial	30	306	7365	220920	2651040
Residential	2500	2350	56675	1700000	20400000
Industrial	10	706	16940	508180	6098160
Transportation	5	8.6	206.4	6192	74304
Public/Infrastructure	18	7812	187488	5624640	68415120

(References: https://findenergy.com/nj/glassboro-electricity/ https://docs.google.com/spreadsheets/d/1vYxSWVCP97nBmEsHMBNuPSw1WBdYjK9Ob2S4msdCa74/edit?usp=sharing)

Average Energy Consumption in Glassboro:

(Hypothetical # of Installations)

Commercial (30):

10.2 kWh/hour = 245.5 kWh/day = 7364 kWh/month = 88368 kWh/year

(*30) = 306 kWh/hr ; 7365 kWh/day ; 220920 kWh/month ; 2651040 kWh/yr

Residential (2500):

0.94 kWh/hour = 22.67 kWh/day = 680 kWh/month = 8160 kWh/year

(*2500) = 2350 kWh/hr ; 56675 kWh/day ; 1700000 kWh/month ; 20400000 kWh/yr

(Reference: https://findenergy.com/nj/glassboro-electricity/)

Average Energy Consumption in Glassboro:

(Hypothetical # of Installations)

Industrial (10):

70.6 kWh/hour = 1694 kWh/day = 50818 kWh/month = 609816 kWh/year

(*10) = 706 kWh/hr ; 16940 kWh/day ; 508180 kWh/month ; 6098160 kWh/yr

Transportation (5):

1.72 kWh/hour = 41.28 kWh/day = 1238.4 kWh/month = 14860.8 kWh/year

(*5) = 8.6 kWh/hr ; 206.4 kWh/day ; 6192 kWh/month ; 74304 kWh/year

Public/Infrastructure (18):

434 kWh/hour = 10416 kWh/day = 312480 kWh/month = 3800840 kWh/year

(*18) = 7812 kWh/hr ; 187488 kWh/day ; 5624640 kWh/month ; 68415120 kWh/yr

(Reference: https://findenergy.com/nj/glassboro-electricity/)

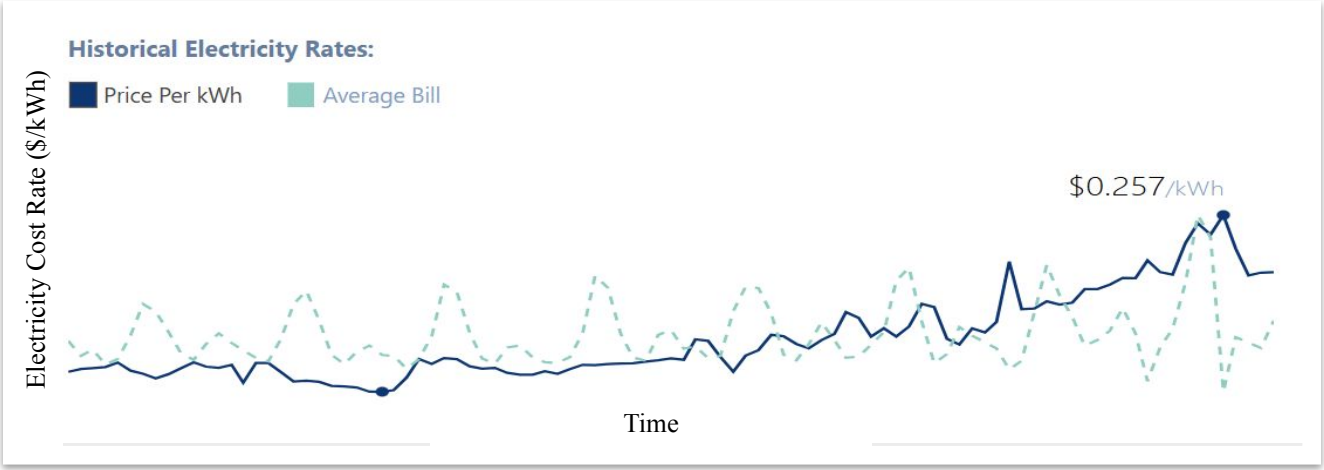


Figure 5: Average Residential Energy Consumption

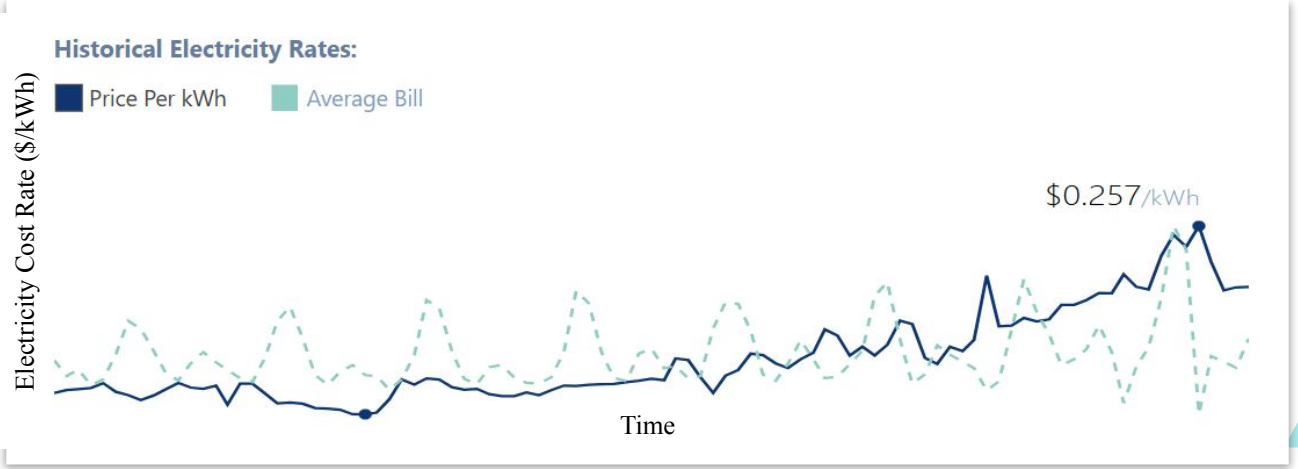


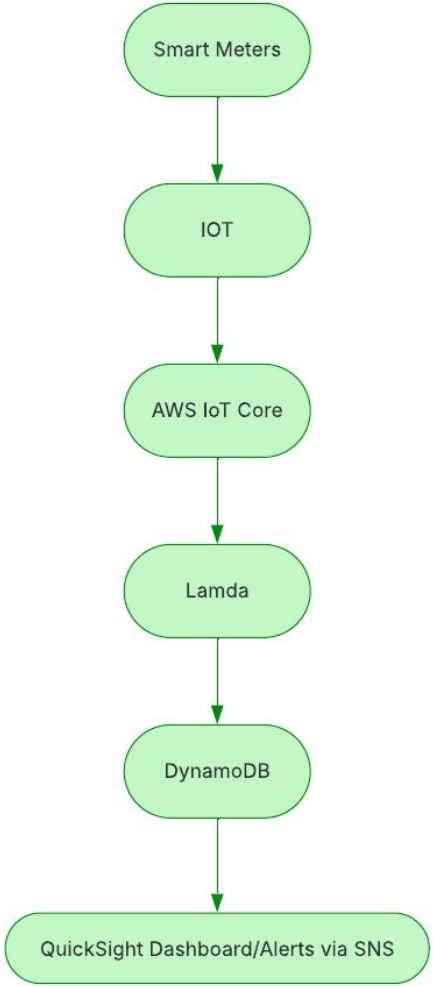
Figure 6: Average Commercial Energy Consumption

(Reference: https://findenergy.com/nj/glassboro-electricity/)

Solution using IoT

IoT (Internet of Things)

- A network of physical objects equipped with sensors & technologies that enable connectivity and data exchange via the internet.
- **Examples:**
  - Smart home devices (thermostats, lighting systems, and voice assistants, wearable devices)
- **Benefits:**
  - Improved traffic management and waste collection
  - Enhance efficiency
  - Provides real-time data, improving convenience & safety



Reference: https://pmc.ncbi.nlm.nih.gov/articles/PMC9667851/

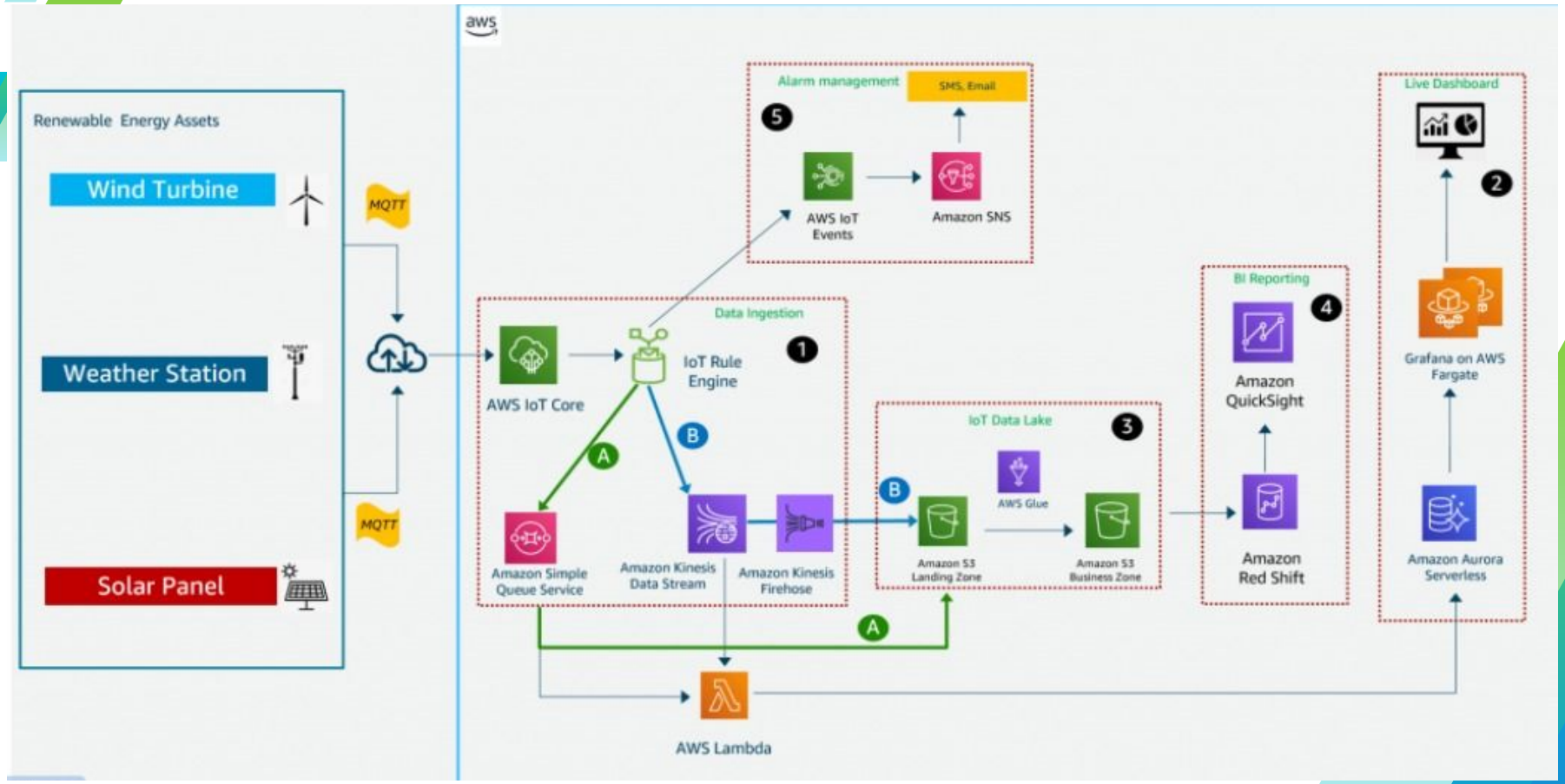
Proposed Solution

- A smart microgrid framework will be established to ensure a reliable energy demand management system, minimize reliance on centralized grids, and improve power generation and energy usage.
- Use of IoT devices is critical to achieve the above objectives.
  - Real-time IoT devices
    - Smart meters, sensors
- **Goal:** Allow for efficient load balancing and the lessening of peak demand pressures.

AWS IoT

AWS IoT Definition:
Cloud service that is able to connect internet with devices via AWS software.

- Smart devices can subscribe to be notified of alerts on state changes – these ‘alerts’ are sent via the message broker (AWS software)



Reference: https://us-east-1.console.aws.amazon.com/iot/home?region=us-east-1#/home

Future Work

Current Challenges:

- Data Security
- Device Connectivity

Potential Solutions:

- Use AWS IAM, encryption
- Use MQTT protocol, edge computing

Future Work:

- Improve data security using advanced monitoring and regular audits.
- Enhance device connectivity by expanding edge computing capabilities.

Conclusion

- The Smart Energy System for Glassboro offers a sustainable and efficient solution by integrating: renewables, smart grids, & real-time monitoring
- Benefits include reduced energy costs & emissions while supporting future growth.
- With AI forecasting and the further integration of technology (ex: Evs), it can enhance Glassboro’s path to a greener future.

Thank You

Any Questions?